Elements of Probability and Statistics

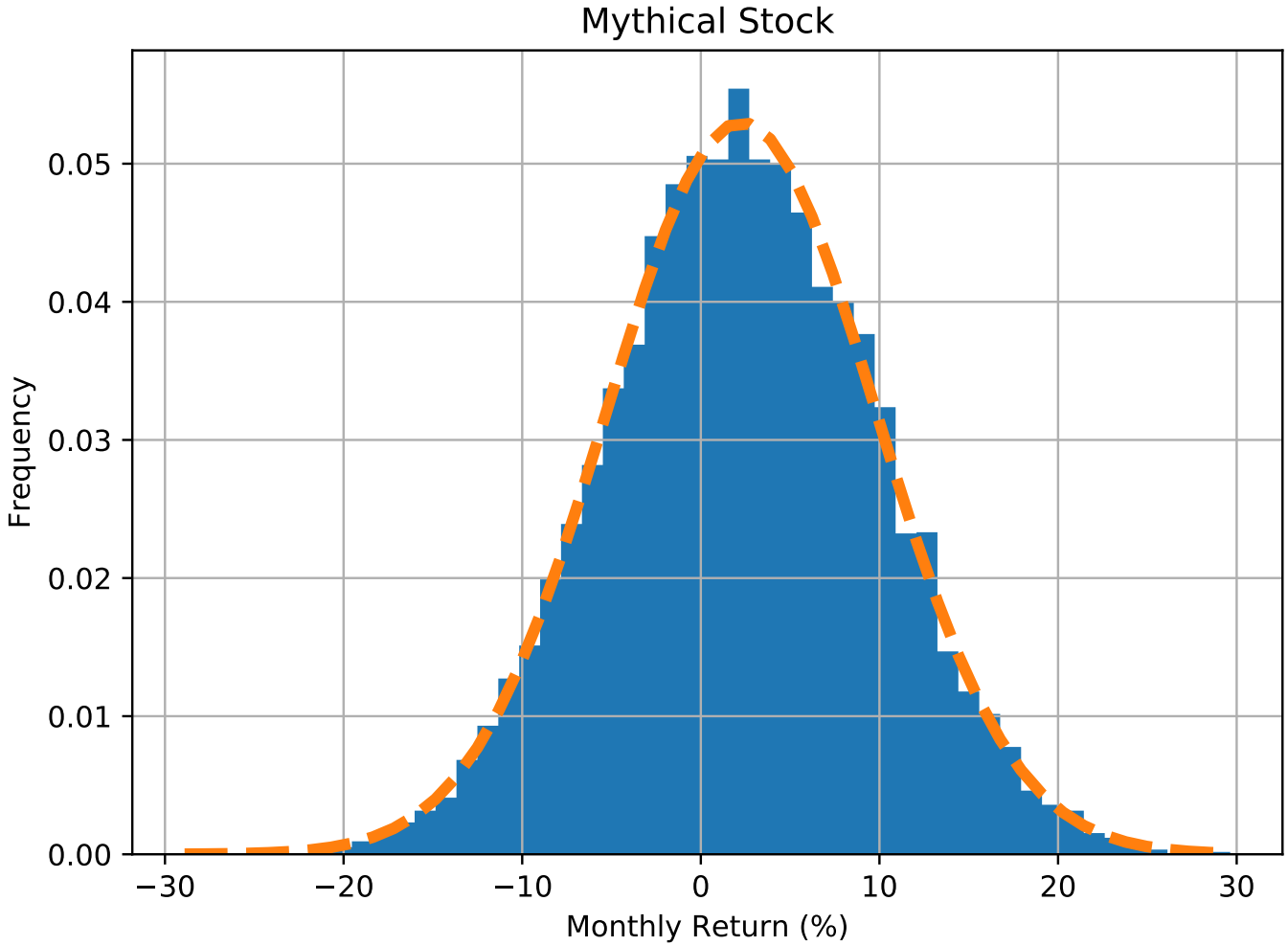
February 26, 2020

- Probability and Statistics are not the same
- A probability is a number
- E.g. we believe that a coin is heads with probability 0.5 and tails with probability 0.5
- A statistic is an *estimation* of a number
- E.g. we flip a coin 10 times – 4 heads, 6 tails
- Our estimate is that the coin is heads with probability 0.4 and tails with probability 0.6

The Normal Distribution

- Assumption: returns are normally distributed
- Consequently, the returns of a portfolio are also normally distributed
- The returns of a portfolio are characterized by three attributes of its constituent stocks:
  1. the mean return of each stock;
  2. the standard deviation (volatility) of each stock;
  3. the covariance between each pair of stocks.
- We will let $R_{s,t}$ denote the random return of stock s on date t
- We will focus on monthly returns, although you can just as well think of hourly, daily, weekly, monthly, yearly returns

The Normal Distribution



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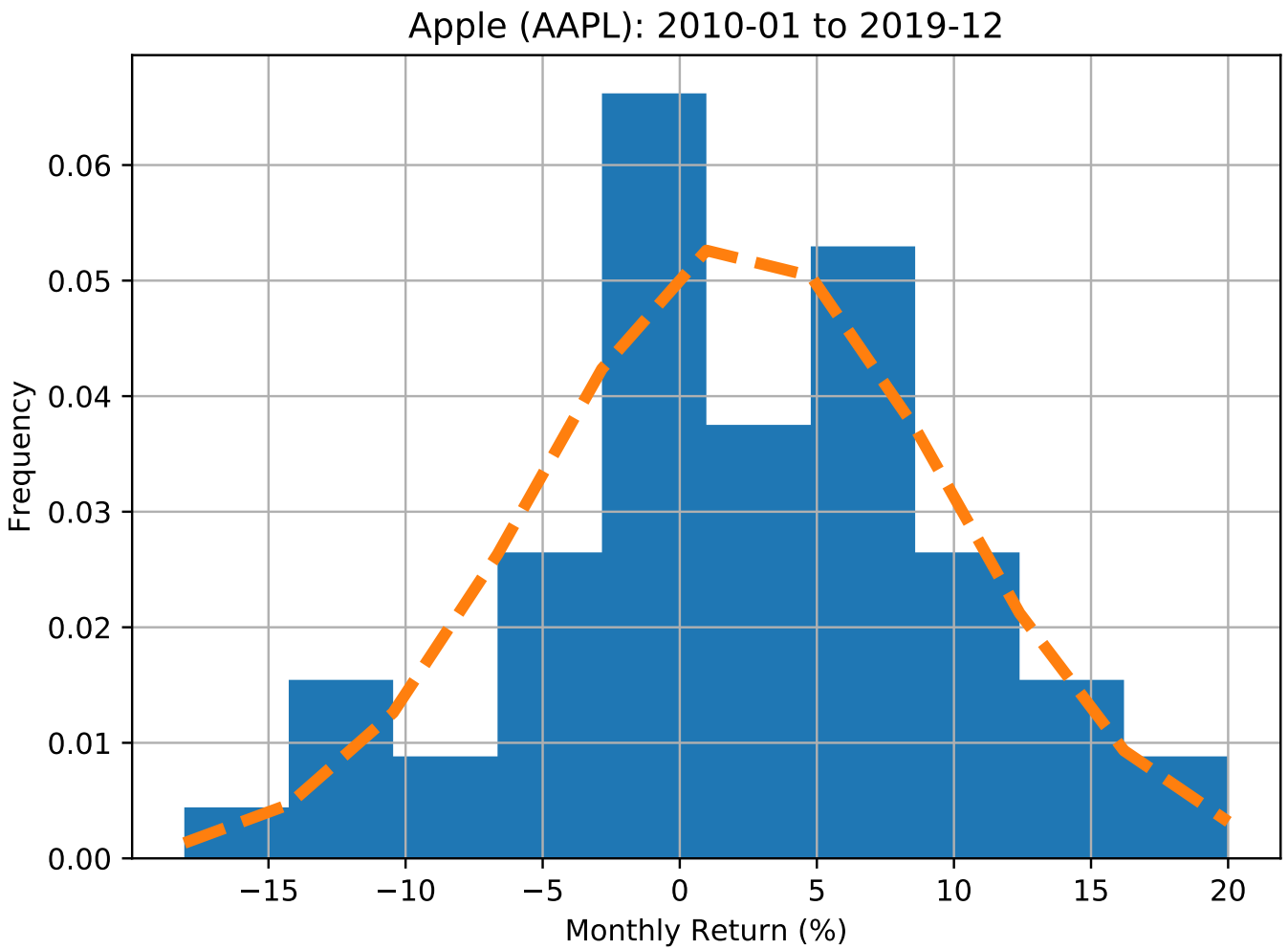

Mean

- We denoted by $\mathrm{E}[R_s]$ the mean, or expectation of R_s
- It is just a <u>number</u>
- It tells what the typical return
- The mean is expressed in "%"
- Our <u>estimate</u> of the mean return for stock s is

$$\mu_s = \frac{1}{N} \sum_{t=1}^{N} R_{s,t}$$

where $R_{s,t}$ are the <u>realized</u> returns

Estimating the Mean

What are your estimates of the mean returns for Apple (AAPL) and Home Depot (HD) given the following data?

date	AAPL return	HD return
2010-01-01	-8.86%	-3.18%
2010-02-01	6.54%	11.39%
2010-03-01	14.85%	4.44%

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$$\mu_{\text{AAPL}} = \frac{-8.86\% + 6.54\% + 14.85\%}{3} = 4.1767\%$$

$$\mu_{\text{HD}} = \frac{-3.18\% + 11.39\% + 4.44\%}{3} = 4.2167\%$$

Variance

- We denoted by $\mathrm{Var}[R_s]$ the variance of R_s
- The variance is used to construct the standard deviation
- The variance does <u>not</u> have units
- Our <u>estimate</u> of the variance of stock s is

$$\sigma_s^2 = \frac{1}{N-1} \sum_{t=1}^{N} (R_{s,t} - \mu_s)^2$$

- The "$N-1$" makes the estimator unbiased

Estimating the Variance

What is your estimate of the variance of the return for Apple (AAPL) given the data?

Estimating the Variance

What is your estimate of the variance of the return for Apple (AAPL) given the data? We already computed the mean: 4.1767%. Don't use % in you calculation!

$$\sigma^2_{\text{AAPL}} = \frac{1}{2}\left[(-0.0886 - 0.041767)^2 + (0.0654 - 0.041767)^2 + (0.1485 - 0.041767)^2\right] = 0.014473$$

Estimating the Variance

What is your estimate of the variance of the return for Home Depot (HD) given the data?

Estimating the Variance

What is your estimate of the variance of the return for Home Depot (HD) given the data? We already computed the mean: 4.2167%. Don't use % in you calculation!

$$\sigma^2_{\text{HD}} = \frac{1}{2}\left[(-0.0318 - 0.042167)^2 + (0.1139 - 0.042167)^2 + (0.0444 - 0.042167)^2\right] = 0.005311$$

Standard Deviation

- We denote by $\mathrm{SD}[R_s]$ the standard deviation or volatility of R_s
- It is our measure of risk
- The standard deviation is $\mathrm{SD}[R_s] = \sqrt{\mathrm{Var}[R_s]}$
- It is expressed in "%"

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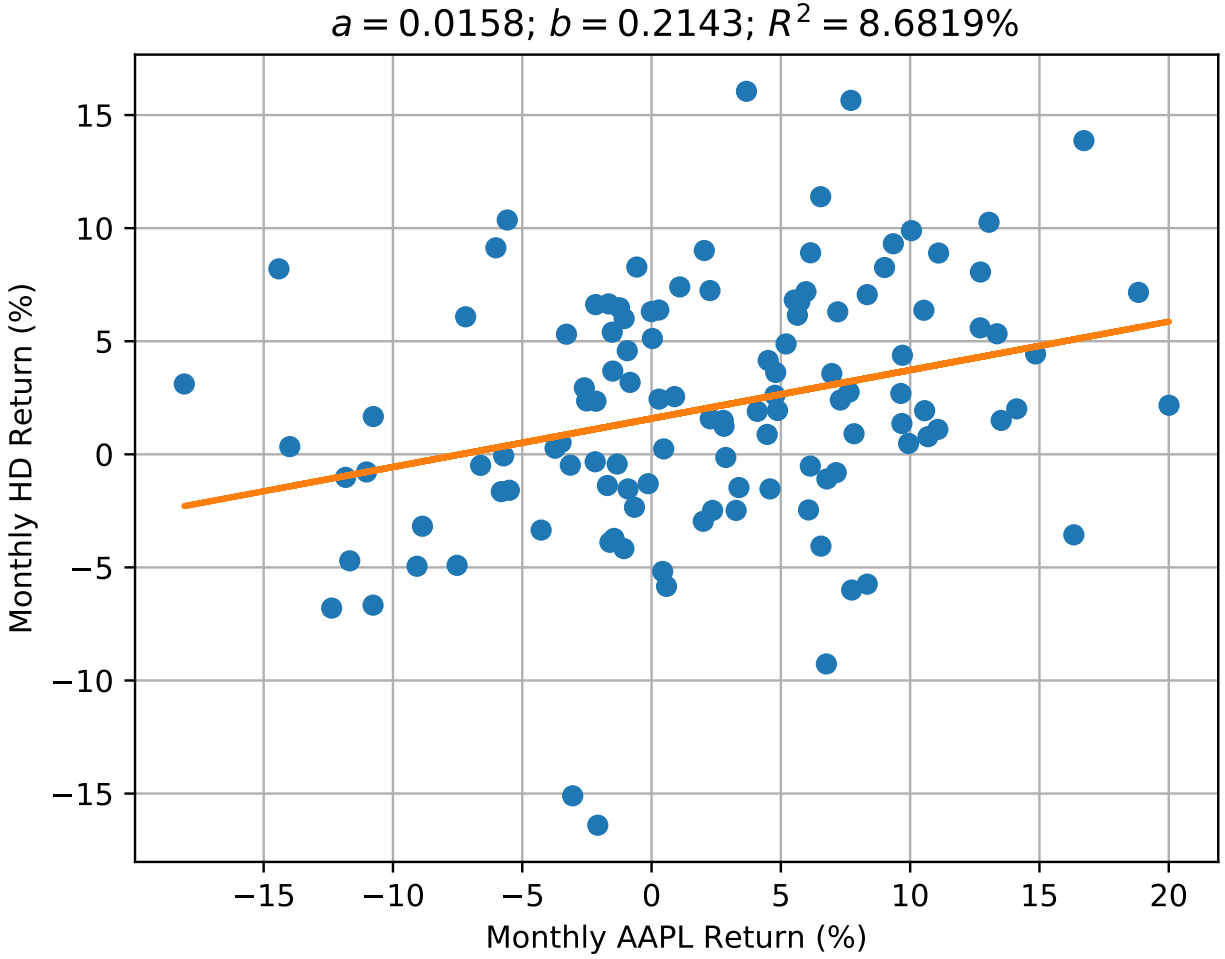
- It is expressed in "%"
- Our estimate in the Apple (AAPL) example is $12.0304\% = \sqrt{0.014473}$
- Our estimate in the Home Depot (HD) example is $7.2877\% = \sqrt{0.005311}$


$a = 0.0158$; $b = 0.2143$; $R^2 = 8.6819\%$
Monthly HD Return (%)
Monthly AAPL Return (%)
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
15
20

Covariance

- Let's think about two stocks, A and B
- We denote by $\mathrm{Cov}[R_A, R_B]$ the covariance between the returns on A and B
- The covariance is used to construct the correlation
- The covariance does <u>not</u> have units
- Our <u>estimate</u> of the covariance is

$$\sigma_{A,B} = \frac{1}{N-1} \sum_{t=1}^{N} (R_{A,t} - \mu_A)(R_{B,t} - \mu_B)$$

Estimating the Covariance

What is your estimate of the covariance of the returns for Apple and Home Depot given the data?

date	AAPL return	HD return
2010-01-01	-8.86%	-3.18%
2010-02-01	6.54%	11.39%
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Estimating the Covariance

What is your estimate of the covariance of the returns for Apple and Home Depot given the data?

Estimating the Covariance

What is your estimate of the covariance of the returns for Apple and Home Depot given the data? We already computed the mean: 4.1767%. Don't use % in you calculation!

$$\begin{aligned}\sigma_{\text{AAPL,HD}} = \frac{1}{2}\,[&(-0.0886 - 0.041767)(-0.0318 - 0.042167)\\ &+(0.0654 - 0.041767)(0.1139 - 0.042167)\\ &+(0.1485 - 0.041767)(0.0444 - 0.042167)]\\ &= 0.005788\end{aligned}$$

Correlation

- We denote by $\mathrm{Corr}[R_A, R_B]$ the correlation between the returns of A and B
- It is given by

$$\mathrm{Corr}[R_A, R_B] = \frac{\mathrm{Cov}[R_A, R_B]}{\mathrm{SD}[R_A]\mathrm{SD}[R_B]}$$

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Our estimate in the Apple, Home Depot example is

$$\begin{aligned}\text{Corr}[R_{AAPL}, R_{HD}] &= \frac{\text{Cov}[R_{AAPL}, R_{HD}]}{\text{SD}[R_{AAPL}]\text{SD}[R_{HD}]} \\ &= \frac{0.005788}{.0120304 \times 0.072877} = 0.6602\end{aligned}$$

Correlation

The correlation is between -1 and $+1$, with a correlation of -1 meaning perfectly negatively correlated (A goes up, B goes down), a correlation of 0 meaning they are not correlated at all (A's return doesn't tell us anything about B's return), and a correlation of $+1$ meaning perfectly positively correlated (A goes up, B goes up)

Portfolio of Two Stocks

- We are going to focus on portfolios with two stocks
- All of the computations that follow can be done with an arbitrary number of stocks — we just need some linear algebra
- Consider a portfolio with x_A in stock A and x_B in stock B
- x_A and x_B must sum to one: $x_A + x_B = 1$
- $x > 1$ means we are borrowing (or financing with short)
- $x < 0$ means we are shorting
- The random return of the portfolio is

$$R_P = x_A R_A + x_B R_B$$

Mean and Volatility

- The mean return of the portfolio is

$$\mathrm{E}[R_P] = x_A \mathrm{E}[R_A] + x_B \mathrm{E}[R_B]$$

- The variance of the portfolio is

$$\mathrm{Var}[R_P] = x_A^2 \mathrm{Var}[R_A] + x_B^2 \mathrm{Var}[R_B] + 2 x_A x_B \mathrm{Cov}[R_A, R_B]$$

- Therefore, the volatility of the portfolio is

$$\mathrm{SD}[R_P] = \sqrt{\mathrm{Var}[R_P]}$$

Mean and Volatility

What is the mean return and volatility of a portfolio that is 25% in Apple (AAPL) and 75% in Home Depot (HD)?

Mean and Volatility

What is the mean return and volatility of a portfolio that is 25% in Apple (AAPL) and 75% in Home Depot (HD)? The mean return is

$$\mu_P = .25 \times (4.1767\%) + .75 \times (4.2167\%) = 4.2067\%$$

The variance is

$$\begin{aligned}\sigma_P^2 &= (.25)^2(0.014473) + (.75)^2(0.005311) + 2(.25)(.75)(0.005788) \\ &= 0.006063\end{aligned}$$

and hence the volatlity is

$$\sigma_P = \sqrt{0.006063} = 7.7862\% \tag{1}$$

Intercept and Slope

- Suppose that we regress the returns of stock B on the returns of stock A (the ordering is important)
- Put differently, A is our x-axis and B is our y-axis
- We want to choose a and b to "fit" the equation

$$R_B = a + bR_A \tag{2}$$

  to the data
- The estimates for a and b are

$$\hat{b} = \frac{\mathrm{Cov}[R_A, R_B]}{\mathrm{Var}[R_A]} = \frac{\mathrm{SD}[R_B]}{\mathrm{SD}[R_A]}\mathrm{Corr}[R_A, R_B]$$
$$\hat{a} = \mathrm{E}[R_B] - \hat{b}\mathrm{E}[R_A]$$

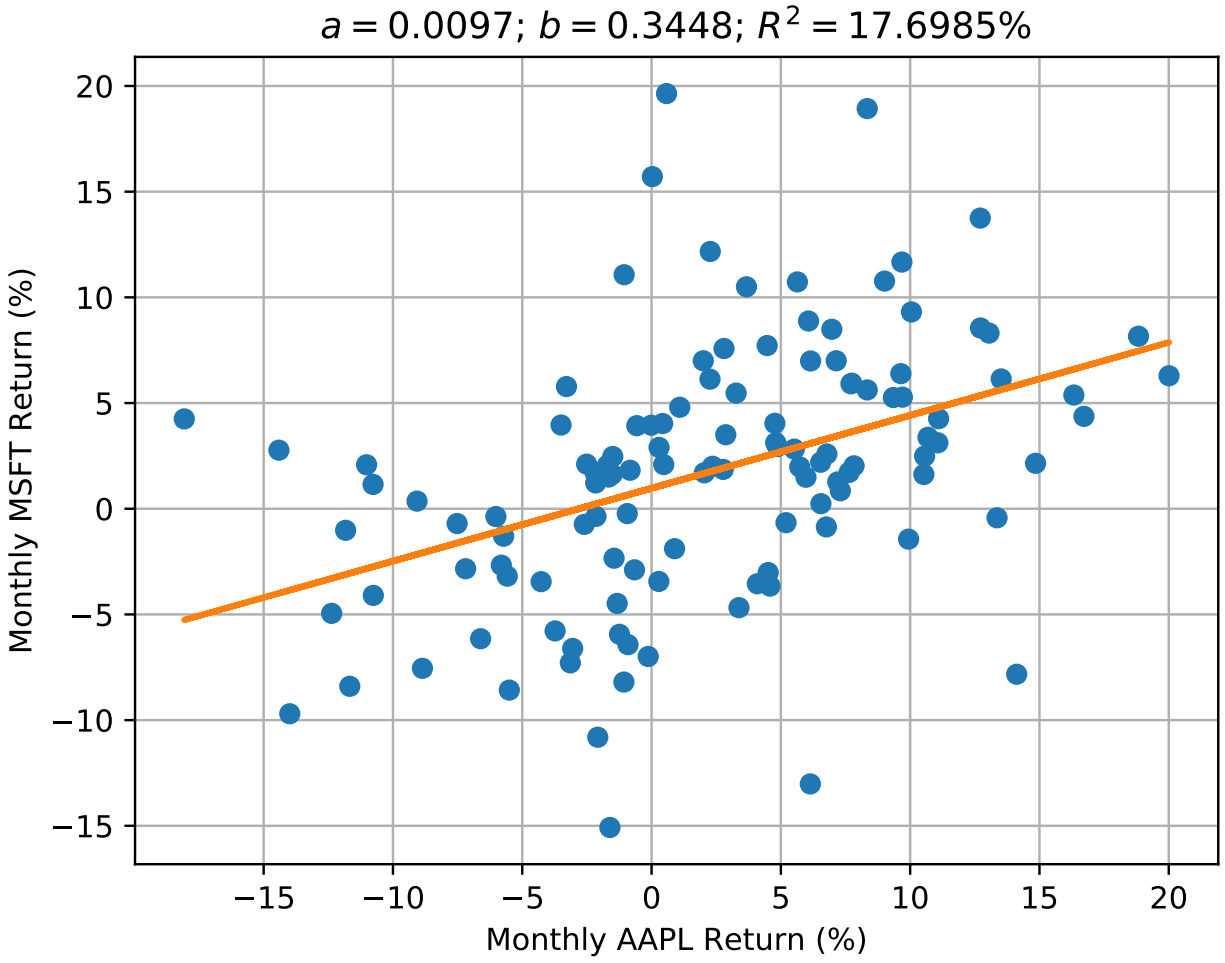
R-Squared

- The *R-squared* (written as R^2) is a measure of the "goodness-of-fit"
- It is the ratio of the variance in the data that can be explained by the regression line to the total variance in the data:

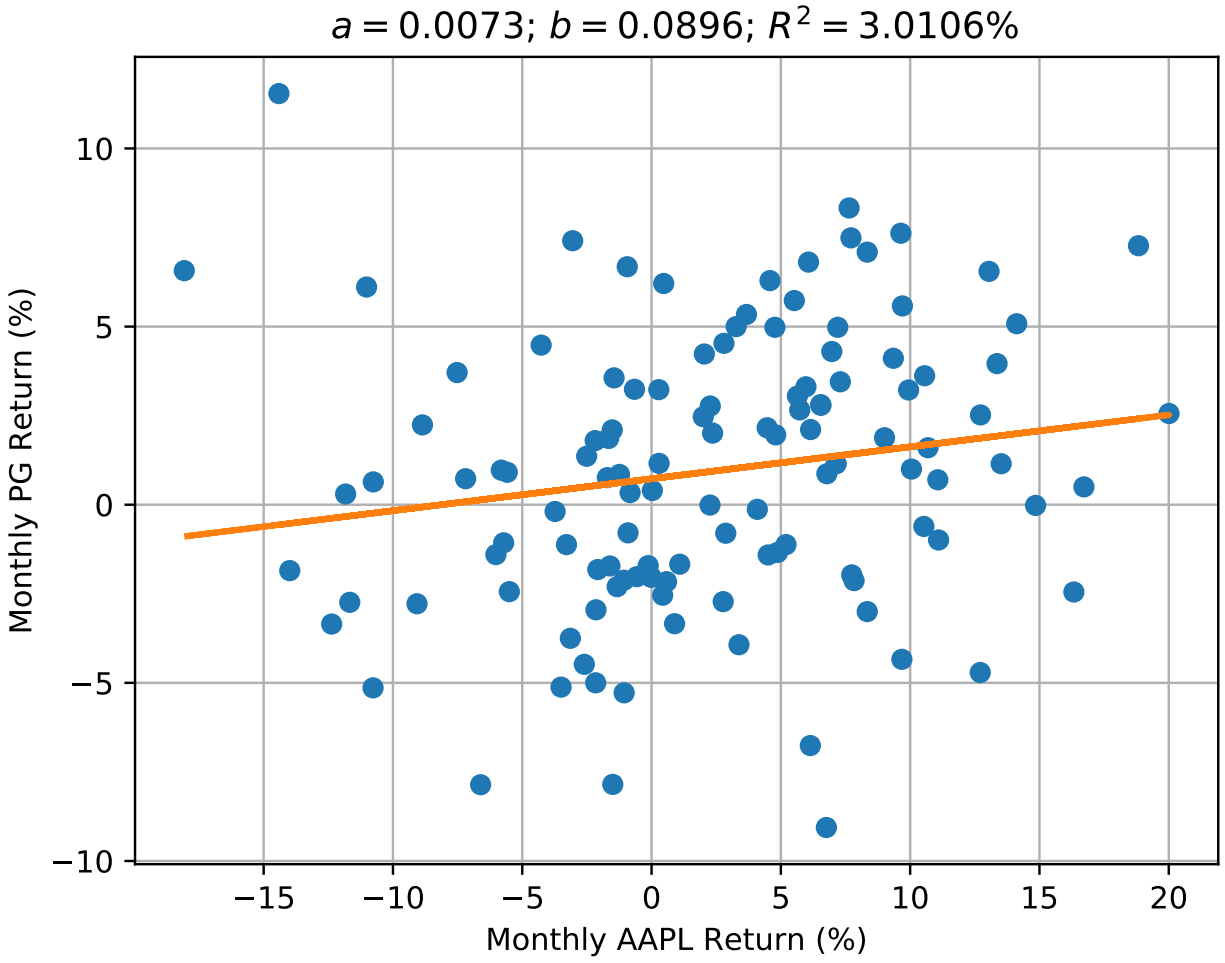
$$R^2 = \frac{b^2 \mathrm{Var}[R_A]}{\mathrm{Var}[R_B]} = \mathrm{Corr}[R_P]^2 \tag{3}$$

- This is only true with a slope and an intercept; R^2 is a bit more complicated in more general regresssions
- It <u>can</u> be written in terms of %
- An R^2 of 100% means the regression perfectly fits the data.

Apple and MSFT



Apple and P&G


$a = 0.0073$; $b = 0.0896$; $R^2 = 3.0106\%$
Monthly PG Return (%)
Monthly AAPL Return (%)
10
5
0
−5
−10
−15
−10
−5
0
5
10
15
20

Example

Consider American Express (AXP) and Visa (V). Suppose that we regress Visa's returns on those of American Express.
$\mathrm{E}[R_{\mathrm{AXP}}] = 1.2100\%$ $\mathrm{E}[R_{\mathrm{V}}] = 1.9960\%$, $\mathrm{SD}[R_{\mathrm{AXP}}] = 5.7304\%$
$\mathrm{SD}[R_{\mathrm{V}}] = 5.0389\%$, $\mathrm{Corr}[R_{\mathrm{AXP}}] = 0.5150$

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$\mathrm{SD}[R_{\mathrm{V}}] = 5.0389\%$, $\mathrm{Corr}[R_{\mathrm{AXP}}] = 0.5150$

$$\hat{b} = \frac{5.0389\%}{5.7304\%} \times 0.5150 = 0.4529$$

$$\hat{a} = 1.9960\% - 0.4529 \times 1.2100\% = 1.4480\%$$

$$R^2 = (0.5150)^2 = 26.5225\%$$